

EE660 Lecture Notes: Gaussian Process Regression

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1 Problem Setup

We take a short digression from classification to *regression*, where the labels are real-valued. Our previous study of kernel methods allows us to discuss *Gaussian Process (GP) regression*, a powerful tool for regression with *uncertainty quantification*. Given training data $S_n = \{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbf{X} \subseteq \mathbb{R}^d$ and $y_i \in \mathbb{R}$, we would like to predict the *distribution* of the response at a new input $x \in \mathbf{X}$, i.e., $P(f(x) \mid x, S_n)$. We model the observations as

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad (1.1)$$

where $f : \mathbf{X} \mapsto \mathbb{R}$ is an unknown function and $\sigma^2 > 0$.

1.1 A prior over functions

We take a Bayesian approach, which places a *prior distribution* on the unknown function f , and updates this distribution after observing the data S_n :

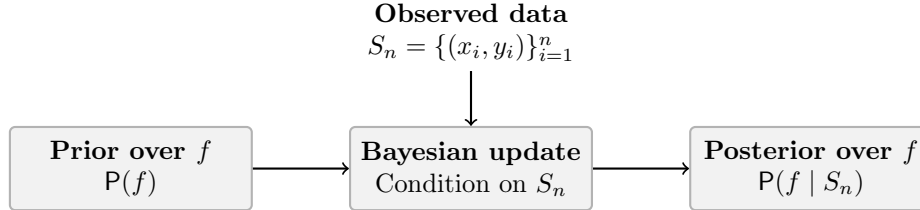


Figure 1: Bayesian learning updates a prior distribution over functions using the observed dataset to obtain a posterior distribution over functions.

In general, it is challenging to even specify what a valid prior distribution over *function spaces* should be. Fortunately, there is one elegant approach via a Gaussian process. A Gaussian process specifies this prior through the joint distribution of the function values at every *finite collection* of inputs.

Definition 1.1 (Gaussian process). Let $m : \mathbf{X} \mapsto \mathbb{R}$ and let $\kappa : \mathbf{X} \times \mathbf{X} \mapsto \mathbb{R}$ be a symmetric positive-semidefinite kernel. We write $f \sim \mathcal{GP}(m, \kappa)$ if, for every $r \geq 1$ and every $Z = (z_1, \dots, z_r) \in \mathbf{X}^r$,

$$(f(z_1), \dots, f(z_r)) \sim \mathcal{N}(m_Z, K_{Z,Z}), \quad (m_Z)_i = m(z_i), \quad (K_{Z,Z})_{ij} = \kappa(z_i, z_j).$$

Here $m(x) = \mathbb{E}[f(x)]$ describes the prior mean and $\kappa(x, x') = \text{Cov}(f(x), f(x'))$ describes how deviations from the mean co-vary at two inputs. A PSD kernel κ means that every Gram matrix $K_{Z,Z}$ is PSD, exactly what is required of a covariance matrix. The word “Gaussian” refers to distributions of *function values*; the inputs need not have a Gaussian distribution, and the kernel need not be a Gaussian kernel. Our complete model is $f \sim \mathcal{GP}(m, \kappa)$ together with (1.1), with the noise independent of f . We regard m , κ , and σ^2 as given. The choice $m \equiv 0$ is common, but keeping a general mean will cost almost no additional work.

2 Gaussian Process Regression Algorithm

We first state the algorithm, before deriving why it gives the conditional distribution under the model above.

2.1 Prediction at a single input

Define the training Gram matrix, prior mean vector, and kernel vector by

$$\begin{aligned} K_{X,X} &= [\kappa(x_i, x_j)]_{i,j=1}^n \in \mathbb{R}^{n \times n}, & m_X &= (m(x_1), \dots, m(x_n)) \in \mathbb{R}^n, \\ k_X(x) &= (\kappa(x_1, x), \dots, \kappa(x_n, x)) \in \mathbb{R}^n, & G_X &= K_{X,X} + \sigma^2 I_n. \end{aligned}$$

Since $K_X \succ 0$ and $\sigma^2 > 0$, G_X is positive definite and therefore invertible.

GP regression training: Form $G_X = K_{X,X} + \sigma^2 I_n$ and compute:

$$\alpha_X = G_X^{-1}(Y - m_X) \in \mathbb{R}^n, \quad Y = (y_1, \dots, y_n) \in \mathbb{R}^n. \quad (2.1)$$

GP regression inference: At a query input $x \in \mathbb{X}$, compute:

$$\mu_n(x) = m(x) + k_X(x)^\top \alpha_X, \quad (2.2)$$

$$s_n^2(x) = \kappa(x, x) - k_X(x)^\top G_X^{-1} k_X(x). \quad (2.3)$$

The posterior distributions is:

$$f(x) \mid (S_n, x) \sim \mathcal{N}(\mu_n(x), s_n^2(x)). \quad (2.4)$$

The point predictor is $\hat{f}_n(x) = \mu_n(x)$. It starts with the prior mean and adds a weighted combination of kernel evaluations:

$$\hat{f}_n(x) = m(x) + \sum_{i=1}^n \alpha_{X,i} \kappa(x_i, x).$$

One big advantage of having access to the posterior distribution $\mathbb{P}(f(x) \mid S_n, x)$ is that it is possible to quantify the *uncertainty* of the prediction at x . For example, we have for every $x \in \mathbb{X}$:

$$\mathbb{P}(f(x) \in [\mu_n(x) - 1.96s_n(x), \mu_n(x) + 1.96s_n(x)]) \approx 0.95.$$

Figure 2 shows a graphical depiction of updating the GP prior to posterior given S_n .

2.2 Joint prediction at multiple inputs

For test inputs $Z = (z_1, \dots, z_q)$, define:

$$\begin{aligned} K_{X,Z} &= [\kappa(x_i, z_j)]_{\substack{1 \leq i \leq n \\ 1 \leq j \leq q}} \in \mathbb{R}^{n \times q}, & K_{Z,Z} &= [\kappa(z_i, z_j)]_{i,j=1}^q \in \mathbb{R}^{q \times q}, \\ K_{Z,X} &= K_{X,Z}^\top \in \mathbb{R}^{q \times n}, & m_Z &= (m(z_1), \dots, m(z_q)) \in \mathbb{R}^q. \end{aligned}$$

The joint posterior is

$$\begin{aligned} (f(z_1), \dots, f(z_q)) \mid (S_n, Z) &\sim \mathcal{N}(\mu_n(Z), \Sigma_n(Z)), \\ \mu_n(Z) &= m_Z + K_{Z,X} \alpha_X, \\ \Sigma_n(Z) &= K_{Z,Z} - K_{Z,X} G_X^{-1} K_{X,Z}. \end{aligned} \quad (2.5)$$

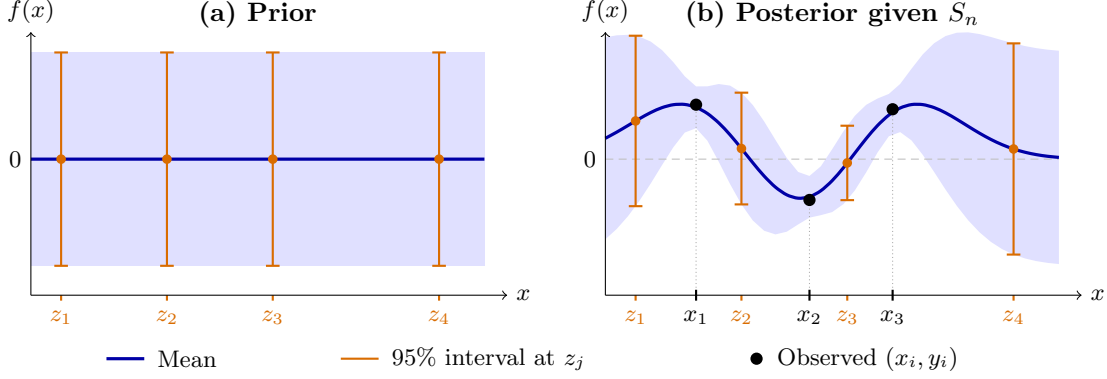


Figure 2: Prior and posterior uncertainty at the same query inputs $z_1, \dots, z_4$. Blue curves show the means; shaded bands and orange bars show pointwise 95% intervals for the latent function f . Conditioning on S_n shifts the mean and reduces uncertainty near the training inputs. Axis labels distinguish test inputs z_j (orange) from training inputs x_i (black). Far from the data, uncertainty remains close to its prior level. Both panels use the same scales.

3 Derivation

The main derivation has two steps: write down a joint Gaussian distribution for the observed and unknown quantities, and then condition on the observations. We first establish the conditioning identity.

3.1 Conditioning a jointly Gaussian vector

The following Gaussian conditioning lemma is the crucial result that we will use to derive the GP regression updates. The same lemma also is used to derive Kalman filters for linear Gaussian state-space models.

Lemma 3.1 (Gaussian conditioning). *Suppose*

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \right), \quad A \succ 0.$$

Then

$$Y \mid X \sim \mathcal{N}(b + B^\top A^{-1}(X - a), D - B^\top A^{-1}B). \quad (3.1)$$

Before we prove Lemma 3.1, we first prove an intermediate result which will motivate the selection of a residual random vector in the proof. Suppose we only observe X , and we want to predict the random variable Y with an affine predictor of the form $b + M(X - a)$. Which M should we choose to minimize the variance of the residual $Y - (b + M(X - a))$?

Proposition 3.2. *Suppose (X, Y) is distributed as in Lemma 3.1. We have:*

$$\arg \min_M \mathbb{E} \| (Y - b) - M(X - a) \|^2 = B^\top A^{-1}. \quad (3.2)$$

Proof. Write $X_a := X - a$ and $Y_b := Y - b$. The covariance assumptions give $\mathbb{E}[X_a X_a^\top] = A$, $\mathbb{E}[X_a Y_b^\top] = B$, and $\mathbb{E}[Y_b Y_b^\top] = D$. Expanding the squared norm and using the cyclic property of the trace yields

$$\begin{aligned} J(M) &:= \mathbb{E} \| Y_b - M X_a \|^2 = \mathbb{E}[Y_b^\top Y_b] - 2\mathbb{E}[Y_b^\top M X_a] + \mathbb{E}[X_a^\top M^\top M X_a] \\ &= \text{tr}(D) - 2\text{tr}(MB) + \text{tr}(MAM^\top). \end{aligned}$$

Using $\nabla_M \text{tr}(MB) = B^\top$ and $\nabla_M \text{tr}(MAM^\top) = M(A + A^\top) = 2MA$, we obtain

$$\nabla J(M) = -2B^\top + 2MA.$$

Setting the gradient equal to zero and using that A is invertible gives

$$\nabla J(M) = 0 \iff MA = B^\top \iff M = B^\top A^{-1}.$$

To verify that this stationary point is the unique minimizer, observe that for every nonzero matrix H of the same dimensions as M ,

$$\left. \frac{d^2}{dt^2} J(M + tH) \right|_{t=0} = 2 \operatorname{tr}(HAH^\top) > 0,$$

where the strict inequality follows from $A \succ 0$. Thus J is strictly convex, and its unique minimizer is $B^\top A^{-1}$. $\square$

Proof of Lemma 3.1. Define the following residual vector R :

$$R = (Y - b) - B^\top A^{-1}(X - a).$$

From Proposition 3.2, the choice of $M = B^\top A^{-1}$ yields an affine predictor $b + M(X - a)$ which has minimal variance over all affine predictors. Furthermore, the pair (X, R) is jointly Gaussian because it is an affine transformation of (X, Y) . Since $\mathbb{E}[X] = a$ and $\mathbb{E}[Y] = b$, we have $\mathbb{E}[R] = 0$. Using the definition of cross-covariance,

$$\begin{aligned} \operatorname{Cov}(X, R) &= \mathbb{E}[(X - a)R^\top] \\ &= \mathbb{E}[(X - a)((Y - b)^\top - (X - a)^\top A^{-1}B)] \\ &= \underbrace{\mathbb{E}[(X - a)(Y - b)^\top]}_B - \underbrace{\mathbb{E}[(X - a)(X - a)^\top]}_A A^{-1}B \\ &= B - AA^{-1}B = 0. \end{aligned}$$

Here we used that A^{-1} is symmetric, since A is a positive-definite covariance matrix. For jointly Gaussian vectors, zero cross-covariance implies independence, and thus $R \perp X$. To compute the covariance of R , write $X_a := X - a$, $Y_b := Y - b$, and $M := B^\top A^{-1}$, so that $R = Y_b - MX_a$. Since $\mathbb{E}[R] = 0$,

$$\begin{aligned} \operatorname{Cov}(R) &= \mathbb{E}[RR^\top] \\ &= \mathbb{E}[(Y_b - MX_a)(Y_b^\top - X_a^\top M^\top)] \\ &= \underbrace{\mathbb{E}[Y_b Y_b^\top]}_D - \underbrace{\mathbb{E}[Y_b X_a^\top]}_{B^\top} M^\top - M \underbrace{\mathbb{E}[X_a Y_b^\top]}_B + M \underbrace{\mathbb{E}[X_a X_a^\top]}_A M^\top. \end{aligned}$$

Using $M^\top = A^{-1}B$ and substituting gives

$$\operatorname{Cov}(R) = D - B^\top A^{-1}B - B^\top A^{-1}B + B^\top A^{-1}AA^{-1}B = D - B^\top A^{-1}B.$$

Finally, write $Y = b + B^\top A^{-1}(X - a) + R$. Given X , the first two terms are fixed and the independent residual R keeps its original Gaussian distribution. This proves (3.1). $\square$

The matrix $D - B^\top A^{-1}B$ is the *Schur complement* of A in the joint covariance; the proof shows why it is positive semidefinite.

3.2 Apply the identity to function values and observations

Let $f_X = (f(x_1), \dots, f(x_n))$. By the Gaussian process prior,

$$\begin{bmatrix} f_X \\ f_Z \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m_X \\ m_Z \end{bmatrix}, \begin{bmatrix} K_{X,X} & K_{X,Z} \\ K_{Z,X} & K_{Z,Z} \end{bmatrix} \right).$$

Now since $Y = f_X + \varepsilon$, $\varepsilon \sim \mathbf{N}(0, \sigma^2 I_n)$ independent of the function, Consequently,

$$\begin{bmatrix} Y \\ f_Z \end{bmatrix} \sim \mathbf{N} \left(\begin{bmatrix} m_X \\ m_Z \end{bmatrix}, \begin{bmatrix} K_{X,X} + \sigma^2 I_n & K_{X,Z} \\ K_{Z,X} & K_{Z,Z} \end{bmatrix} \right). \quad (3.3)$$

Apply Lemma 3.1, taking the lemma's variables (X, Y) to be (Y, f_Z) and setting $A = K_{X,X} + \sigma^2 I_n$, $B = K_{X,Z}$, and $D = K_{Z,Z}$. This gives exactly (2.5). Taking $q = 1$ gives (2.2)–(2.4).

Since this calculation works for every finite set of test inputs, the posterior is itself a Gaussian process:

$$\begin{aligned} f \mid S_n &\sim \mathcal{GP}(\mu_n, \kappa_n), \\ \kappa_n(x, x') &= \kappa(x, x') - k_X(x)^\top (K_{X,X} + \sigma^2 I_n)^{-1} k_X(x'). \end{aligned}$$

In particular, $s_n^2(x) = \kappa_n(x, x)$. No discretization of the entire input space is required: the finite joint distributions suffice to carry out inference at any desired inputs.

4 A second view: Bayesian linear regression in feature space

The GP formulation also extends a familiar finite-dimensional model. Let $\Phi : \mathbf{X} \mapsto \mathbb{R}^D$ be a feature map and consider

$$w \sim \mathbf{N}(0, I_D), \quad f(x) = m(x) + \Phi(x)^\top w. \quad (4.1)$$

Every finite vector of function values is an affine transformation of w and hence Gaussian, with covariance kernel $\kappa(x, x') = \Phi(x)^\top \Phi(x')$. Let $F_n \in \mathbb{R}^{n \times D}$ have rows $\Phi(x_i)^\top$.

We condition on the training inputs X , treating them as fixed, and use the prior $w \mid X \sim \mathbf{N}(0, I_D)$. The observation model is

$$Y = m_X + F_n w + \varepsilon, \quad \varepsilon \mid X \sim \mathbf{N}(0, \sigma^2 I_n),$$

where ε and w are independent conditional on X . Since (Y, w) is an affine transformation of the jointly Gaussian vector (w, ε) , it is jointly Gaussian conditional on X . Its moments are

$$\begin{aligned} \mathbb{E}[Y \mid X] &= m_X, & \mathbb{E}[w \mid X] &= 0, \\ \text{Cov}(Y \mid X) &= F_n F_n^\top + \sigma^2 I_n, & \text{Cov}(w \mid X) &= I_D, \\ \text{Cov}(Y, w \mid X) &= F_n \text{Cov}(w \mid X) + \text{Cov}(\varepsilon, w \mid X) = F_n. \end{aligned}$$

Consequently, defining $G_n := F_n F_n^\top + \sigma^2 I_n$, we have

$$\begin{bmatrix} Y \\ w \end{bmatrix} \mid X \sim \mathbf{N} \left(\begin{bmatrix} m_X \\ 0 \end{bmatrix}, \begin{bmatrix} G_n & F_n \\ F_n^\top & I_D \end{bmatrix} \right). \quad (4.2)$$

Because $\sigma^2 > 0$, $G_n \succ 0$, so we may apply Lemma 3.1. In the lemma, take the first random vector to be Y and the second to be w , with mean vectors $a = m_X$, $b = 0$ and covariance blocks $A = G_n$, $B = F_n$, $D = I_D$. Conditioning on the observed labels gives

$$\begin{aligned} w \mid S_n &\sim \mathbf{N}(\bar{w}_n, \Sigma_n), \\ \bar{w}_n &= F_n^\top G_n^{-1} (Y - m_X), \\ \Sigma_n &= I_D - F_n^\top G_n^{-1} F_n. \end{aligned} \quad (4.3)$$

For a fixed query input x , the function value $f(x) = m(x) + \Phi(x)^\top w$ is an affine transformation of the posterior Gaussian weight vector. Hence,

$$f(x) \mid (S_n, x) \sim \mathbf{N}(m(x) + \Phi(x)^\top \bar{w}_n, \Phi(x)^\top \Sigma_n \Phi(x)). \quad (4.4)$$

To recover the kernel formulas, note that $k_X(x) = F_n^\top \Phi(x)$, $\kappa(x, x) = \Phi(x)^\top \Phi(x)$, and $G_n = K_{X,X} + \sigma^2 I_n$. Substituting the posterior mean and covariance of w gives

$$\begin{aligned} m(x) + \Phi(x)^\top \bar{w}_n &= m(x) + \Phi(x)^\top F_n^\top G_n^{-1} (Y - m_X) \\ &= m(x) + k_X(x)^\top G_n^{-1} (Y - m_X) = \mu_n(x), \\ \Phi(x)^\top \Sigma_n \Phi(x) &= \Phi(x)^\top (I_D - F_n^\top G_n^{-1} F_n) \Phi(x) \\ &= \kappa(x, x) - k_X(x)^\top G_n^{-1} k_X(x) = s_n^2(x). \end{aligned}$$

Thus $f(x) \mid (S_n, x) \sim \mathbf{N}(\mu_n(x), s_n^2(x))$, exactly as in Gaussian process regression.